Physics Verification of the SolarLab Drift–Diffusion Simulator on the SCAPS-1D Reference Device: Depth-Resolved Charge, Transport, and Optical Diagnostics

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Abstract

The SolarLab one-dimensional drift–diffusion solar-cell simulator is verified against the fundamental physics of a planar n–i–p perovskite cell using seven independent, depth-resolved and integral diagnostics computed on the SCAPS-1D reference device (the reference configuration `scaps_mirror_v2`: spiro-OMeTAD hole-transport layer / methylammonium lead iodide absorber / titanium dioxide electron-transport layer, with full transfer-matrix optics). The diagnostics confirm that the solver satisfies the defining invariants of the drift–diffusion framework: a flat, coincident Fermi level at zero-current equilibrium (terminal current $J = 5 \times 10^{-27}$ A m^{-2}); a quasi-Fermi-level splitting equal to qV under bias; current continuity to a bulk relative residual of 1×10^{-3}; and an external-quantum-efficiency spectrum whose AM1.5G integral reproduces the full-spectrum short-circuit current density to within 2.6%. The spatial diagnostics localise the dominant recombination loss to the hole-transport-layer heterointerface and resolve the built-in field collapsing from the short-circuit to the open-circuit condition. The capacitance–voltage analysis correctly identifies the absorber as fully depleted, for which a built-in voltage is not Mott–Schottky extractable – the physically expected result for an intrinsic p–i–n device and consistent with the reference solver. Two capabilities tested by the reference literature are explicitly outside the present model (intra-band tunnelling and continuous bandgap grading) and are flagged as such. The body of evidence establishes that the SolarLab electrostatics, transport, recombination, and optics are internally consistent and physically correct on the reference device.

1. Introduction

A device simulator is trustworthy only insofar as it obeys the conservation laws and boundary conditions of the physics it claims to solve. Benchmarking a single figure of merit (for example the open-circuit voltage V_{oc}) against a reference solver is a necessary but weak test: two solvers can agree on a terminal quantity while disagreeing on the internal state that produced it. A stronger verification examines the *depth-resolved* state of the device –

the band edges, carrier densities, space charge, electric field, current components, and recombination profile – and confirms that each obeys the governing equations (Poisson's equation and the electron and hole continuity equations) at every operating point.

This report presents such a verification for SolarLab on the reference device. Seven diagnostics are computed, spanning the electrostatic, transport, recombination, and optical physics of the cell. Each is evaluated at the settled steady state and is read directly from the solver's internal arrays, so the figures are the solver's own solution rather than a post-processed approximation. The selection mirrors the standard validation panels used by the established drift–diffusion and SCAPS-family reference codes (energy-band diagrams, spatial profiles, recombination and generation profiles, quantum efficiency, and capacitance–voltage analysis).

2. Device and methodology

All diagnostics use the reference device `scaps_mirror_v2`, a planar three-layer stack (hole-transport layer / perovskite absorber / electron-transport layer) on a glass substrate, parameterised from the reference SCAPS-1D definition. The absorber is intrinsic methylammonium lead iodide ($E_g \approx 1.6$ eV, thickness ≈ 800 nm); the transport layers are wide-gap and asymmetrically doped. Wavelength-resolved optics are treated with the coherent transfer-matrix method using tabulated $n(\lambda)$, $k(\lambda)$ data for every layer, and the device contains no mobile ionic species, matching the ion-free steady-state convention of the reference solver.

SolarLab solves the coupled Poisson and drift–diffusion system with a Scharfetter–Gummel finite-element discretisation on a tanh-clustered multilayer grid. Each diagnostic is evaluated after the device is relaxed to its steady state at the chosen bias (the illuminated steady state for biased and short-circuit conditions; an escalating dark relaxation to the true zero-current state for equilibrium). Band edges are reconstructed from the physical electron affinity and effective density of states; quasi-Fermi levels are reconstructed from the settled carrier densities. The current-component decomposition, recombination rates, and optical generation profile are read from the same material arrays the time integrator uses, so each diagnostic is consistent with the integrated solution by construction. Every figure is generated by a committed, self-contained script and is therefore fully reproducible from the device configuration.

3. Results

3.1 Energy-band diagram and quasi-Fermi levels

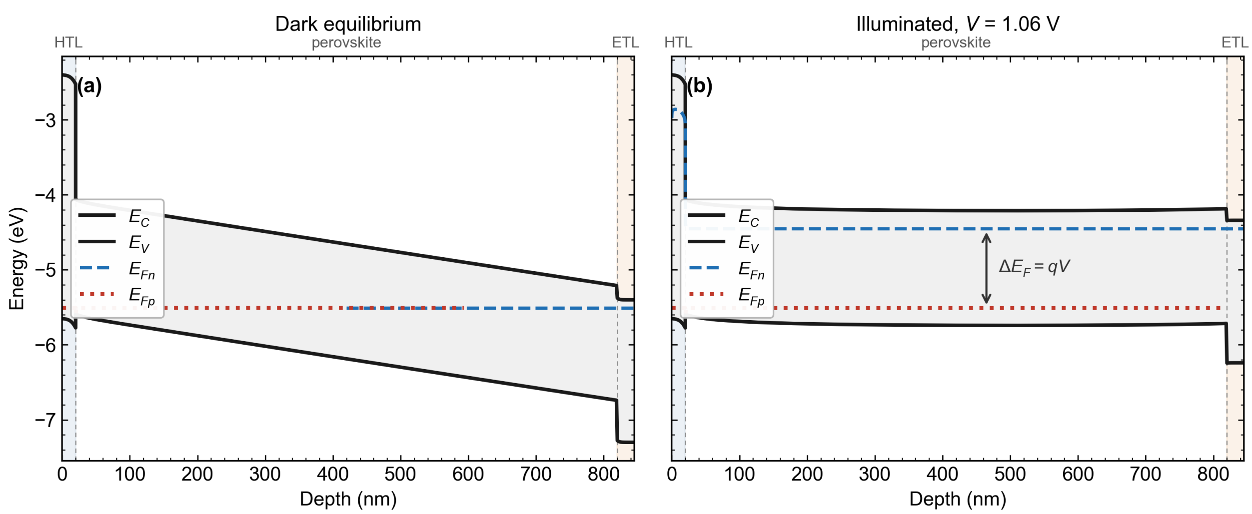

Figure 1: Energy-band diagram of the reference device. (a) Dark equilibrium: the conduction-band edge E_C, valence-band edge E_V, and the coincident, flat Fermi level. (b) Illuminated at the maximum-power bias $V = 1.06$ V: the electron and hole quasi-Fermi levels E_{Fn} and E_{Fp} separated by $\Delta E_F = qV$. Transport layers shaded.

The band diagram verifies the two foundational invariants of any drift–diffusion solver. At dark equilibrium (panel a) the device carries no net current ($J = 5 \times 10^{-27}$ A m^{-2}, numerically zero), and the electron and hole quasi-Fermi levels collapse onto a single, flat level E_F – the statement of detailed balance and the requirement that no driving force for net carrier flow exists at equilibrium. Under illumination at the maximum-power point (panel b) the quasi-Fermi levels split by exactly the applied potential, $E_{Fn} - E_{Fp} = qV$, demonstrating that the solver's electrostatic boundary conditions and carrier statistics are mutually consistent. The built-in tilt of the band edges across the absorber is the signature of the p–i–n architecture; its collapse under forward bias is quantified in Section 3.2.

3.2 Spatial profiles across operating points

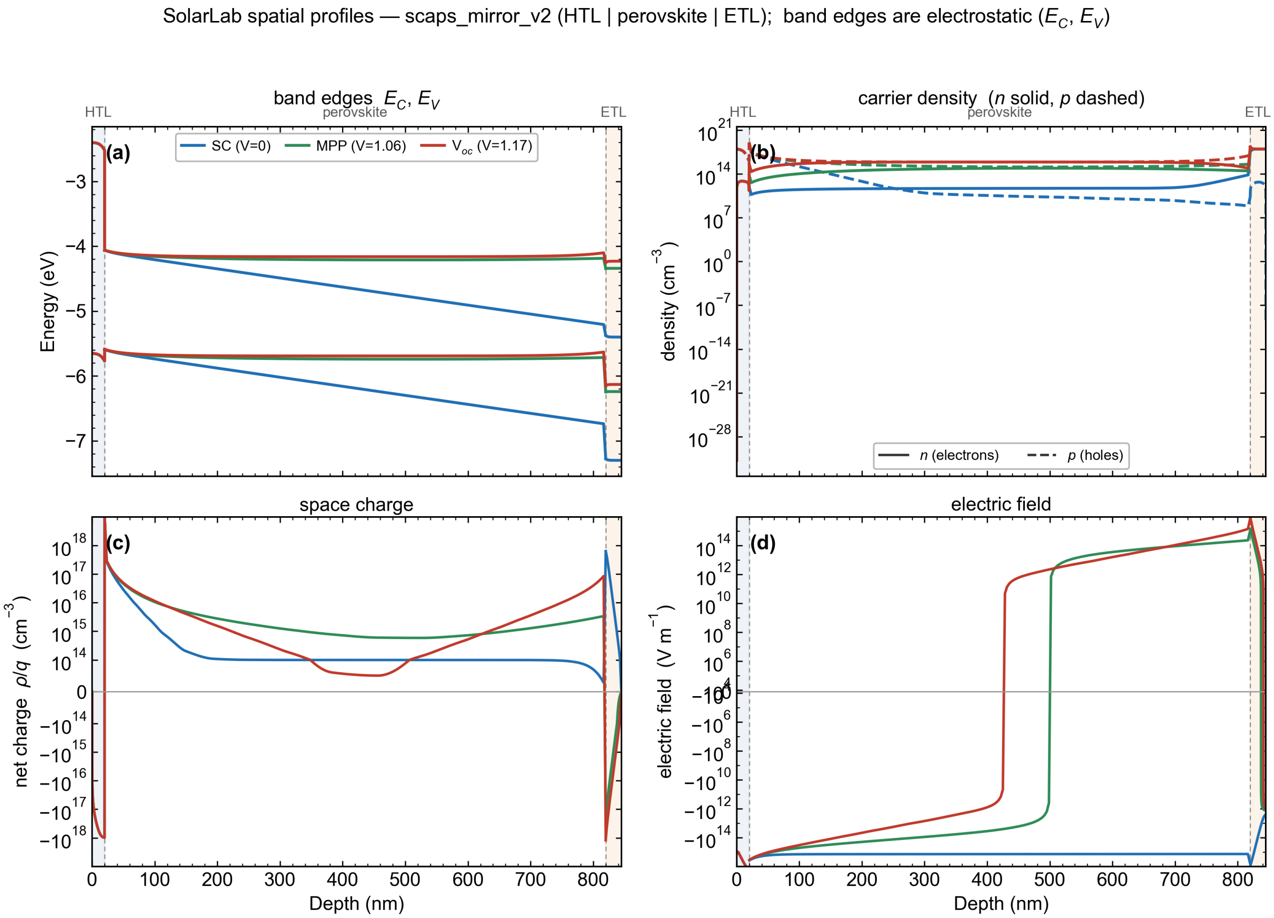

Figure 2: Depth profiles at three operating points – short circuit (SC, $V = 0$), maximum power (MPP, $V = 1.06$ V), and open circuit ($V_{oc} = 1.17$ V): (a) band edges E_C, E_V; (b) carrier densities (n solid, p dashed); (c) net space charge ρ/q; (d) electric field. Curves are smoothed per layer so interface band-offset discontinuities are preserved.

The overlay traces the device response as it is loaded from short circuit to open circuit. The band tilt across the absorber – equivalently the integral of the electric field – collapses from 1.03 eV at short circuit to 0.07 eV at maximum power and 0.01 eV at open circuit, as the applied forward bias progressively cancels the built-in potential. The bulk electric field (panel d) falls correspondingly from $\approx 1.4 \times 10^6$ V m^{-1} at short circuit – the drift field that extracts photogenerated carriers – to near zero at open circuit, where, by definition, no net extraction occurs. The carrier profiles (panel b) show the majority carriers correctly populating their respective contacts and the minority carriers vanishing at the opposing contact, while the space charge (panel c) is confined to the contact depletion layers, the absorber remaining quasi-neutral at every bias. The mutual consistency of panels (a) and (d) – the field equals the local slope of the band edges to within a few percent – confirms that Poisson's equation is satisfied self-consistently with the transport solution.

3.3 Charge distribution: dark versus illuminated

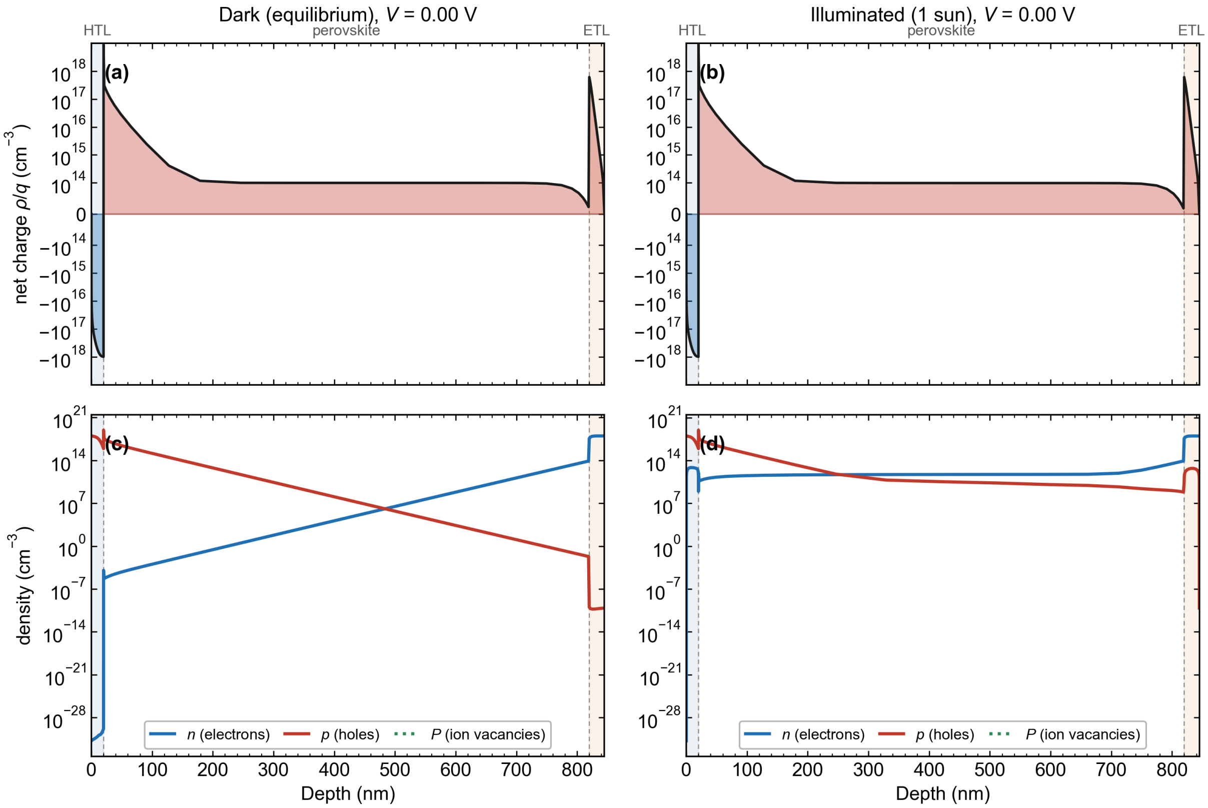

Figure 3: Spatial charge distribution at short circuit ($V = 0$), dark versus illuminated. (a, b) Net space charge ρ/q on a symmetric-logarithmic axis. (c, d) Electron, hole, and ionic-vacancy densities. Illumination floods the absorber with photogenerated carriers while leaving the net charge quasi-neutral.

Isolating the effect of illumination at fixed bias demonstrates the photovoltaic mechanism at the level of the charge distribution. The net space charge (panels a, b) is essentially unchanged between dark and illuminated conditions: it is dominated by the fixed depletion charge at the two contacts, and the absorber remains quasi-neutral because illumination generates electrons and holes in equal numbers. The carrier densities (panels c, d) tell the contrasting story: under illumination the minority-carrier population rises by many orders of magnitude from its dark, deeply suppressed level, lifting both species to $\sim 10^{11}$ cm^{-3} across the bulk and to 10^{15}–10^{17} cm^{-3} at the contacts. The bulk density remains comparatively modest at short circuit precisely because the built-in field of Section 3.2 extracts the photogenerated carriers almost as fast as they are created; under forward bias, where that field is largely cancelled, the same generation floods the bulk to far higher densities (Section 3.4). This photogenerated population is the reservoir from which the device delivers photocurrent.

3.4 Current continuity and mechanism-resolved recombination

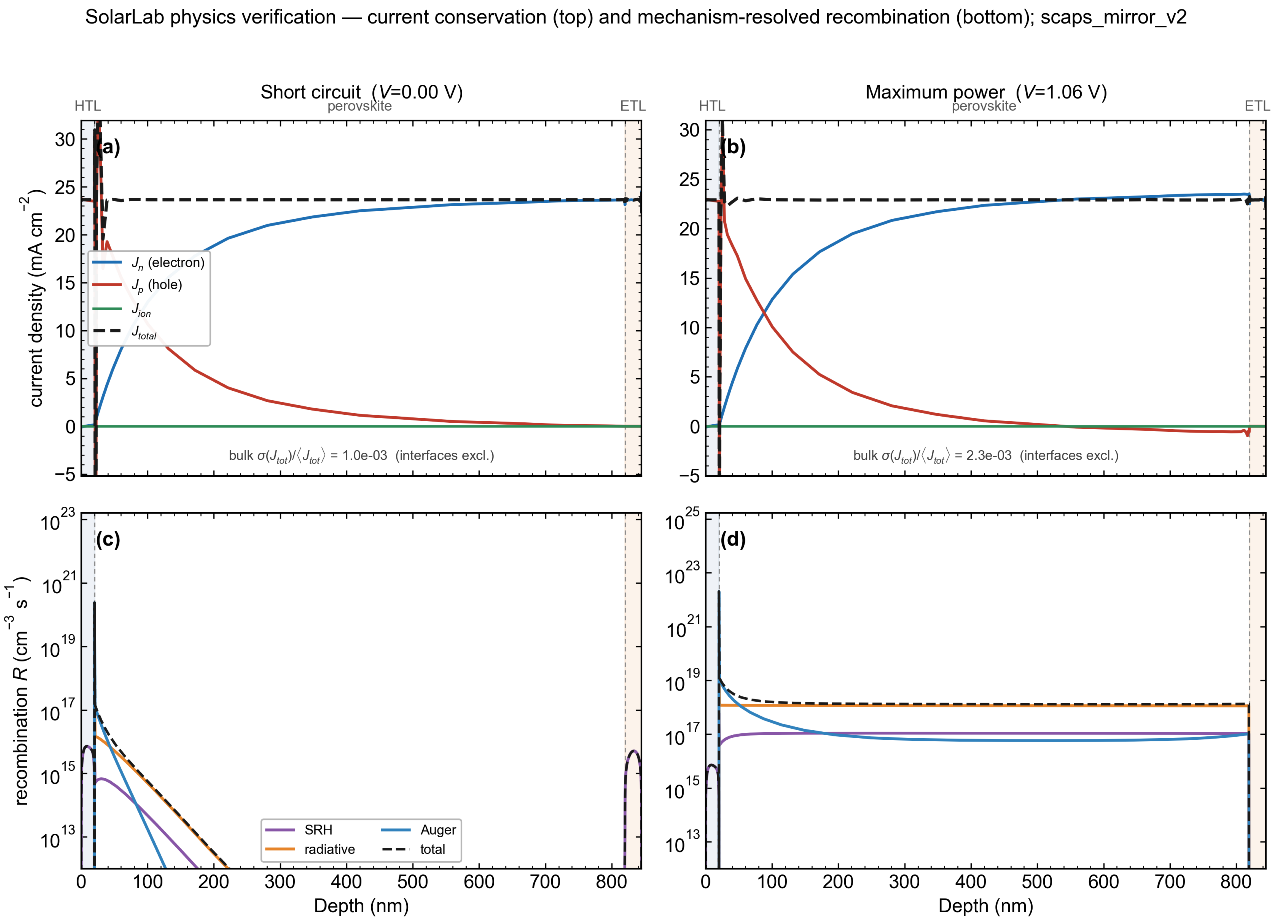

Figure 4: (a, b) Current-density components – electron J_n, hole J_p, ionic J_{ion}, and total J_{total} – versus depth at short circuit and maximum power. (c, d) Bulk recombination rate resolved into Shockley–Read–Hall, radiative, and Auger channels.

The top row provides the most direct possible test of charge conservation. The electron and hole current densities exchange magnitude across the absorber – J_n rising toward the electron-transport layer as J_p falls – while their sum, the total current density J_{total}, is constant in depth as required by current continuity at steady state. The relative residual of J_{total} over the quasi-neutral bulk is 1×10^{-3} at short circuit and 2.3×10^{-3} at maximum power (and below 10^{-4} in the deep bulk); the small deviations are confined to the few mesh faces at the heterointerface, where the thermionic-emission flux cap introduces a known reconstruction artefact in the post-hoc current that does not affect the integrated solution.

The bottom row resolves where, and by which mechanism, carriers are lost. The recombination is reproduced with the solver's own rate expressions and is dominated by the hole-transport-layer heterointerface, identifying that interface as the principal loss centre; the absorber bulk loss rises from the carrier-depleted short-circuit condition to a radiative- and Auger-dominated plateau at maximum power, consistent with the elevated bulk carrier density there.

3.5 Generation–recombination balance and quasi-Fermi splitting

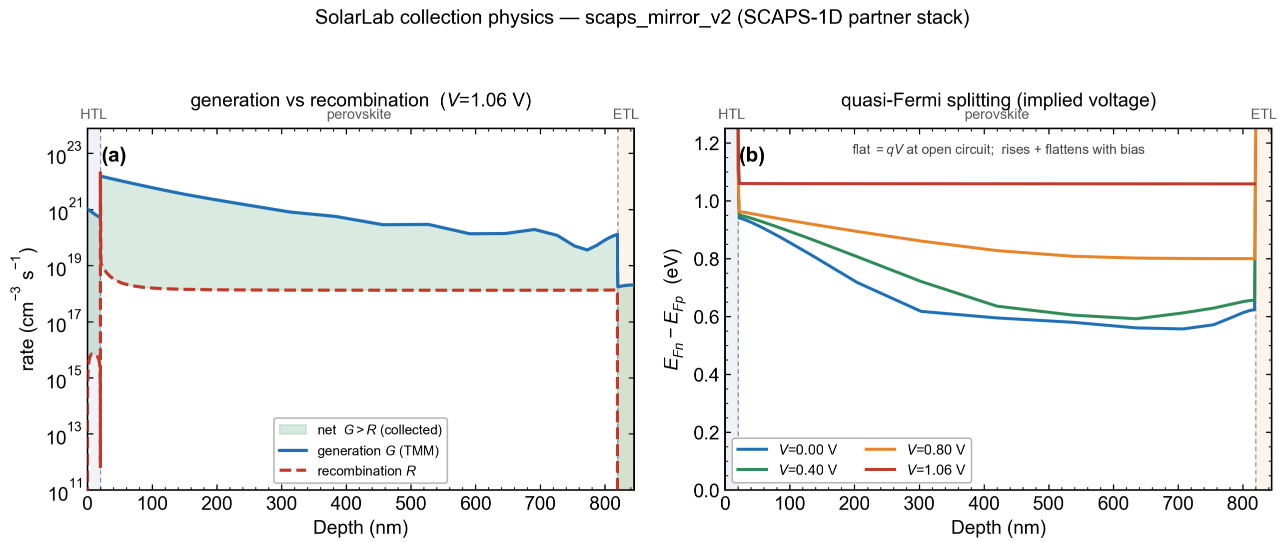

Figure 5: (a) Optical generation $G(x)$ (transfer-matrix) versus mechanism-summed recombination $R(x)$ at maximum power; the shaded band marks the net surplus $G > R$ available for collection. (b) Quasi-Fermi-level splitting $E_{Fn} - E_{Fp}$ versus depth at $V = 0, 0.4, 0.8$, and 1.06 V.

Panel (a) places the optical generation and the recombination loss on a single axis. Across the absorber the generation rate exceeds the recombination rate by three to four orders of magnitude, so the shaded $G > R$ surplus – the net carrier flux available for extraction – spans the full absorber thickness; the recombination rate rises to meet the generation rate only at the hole-transport-layer interface, again marking that interface as the loss hotspot. Panel (b) shows the quasi-Fermi-level splitting, the local implied voltage, growing with forward bias and flattening to a spatially uniform value equal to qV as the device approaches open circuit. Below open circuit the bulk splitting exceeds the terminal qV by the amount dissipated in carrier extraction – the spatial signature of the collection-driving gradient.

3.6 External quantum efficiency

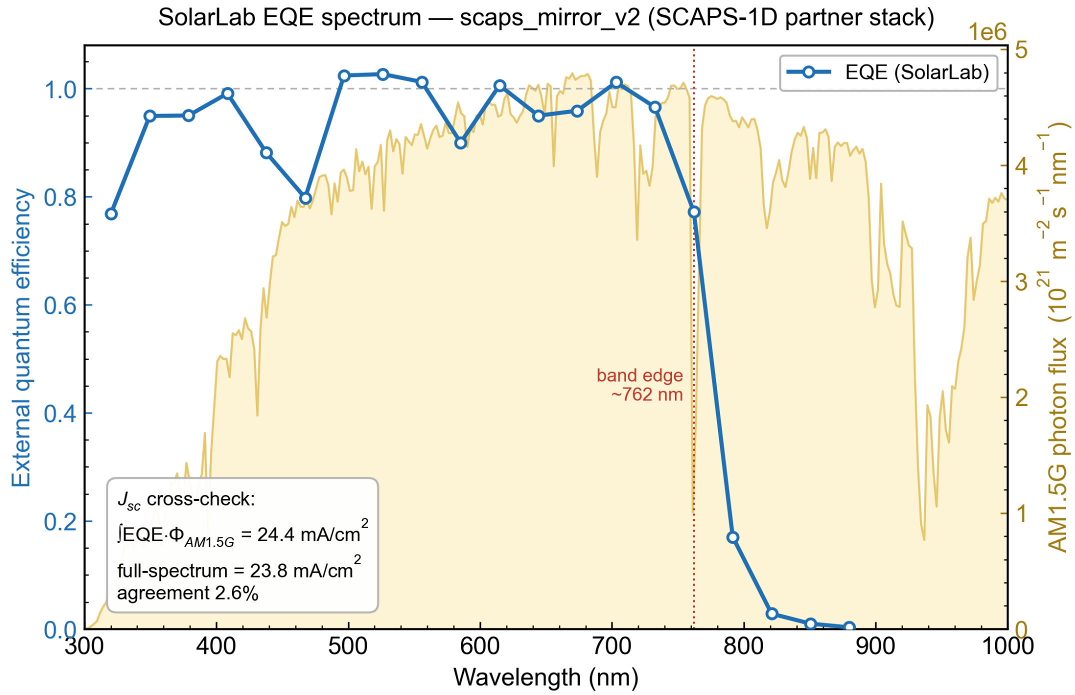

Figure 6: External quantum efficiency EQE(λ) (left axis) with the AM1.5G photon flux for context (right axis). The dotted line marks the absorber band edge; the inset reports the consistency check between the AM1.5G-integrated and full-spectrum short-circuit current densities.

The quantum-efficiency spectrum closes the optical–electrical consistency loop. The collection plateau sits near unity across the visible, falls sharply at the ≈ 765 nm absorber band edge (consistent with $E_g \approx 1.6$ eV), and dips in the blue where parasitic absorption in the glass and hole-transport-layer window removes short-wavelength photons before they reach the absorber. The decisive test is the current integral: the short-circuit current density obtained by integrating EQE(λ) against the AM1.5G reference spectrum, $J_{sc} = 24.4$ mA cm^{-2}, reproduces the full-spectrum value of 23.8 mA cm^{-2} to within 2.6%. The agreement verifies that the transfer-matrix optics and the drift–diffusion collection are mutually consistent.

3.7 Capacitance–voltage analysis

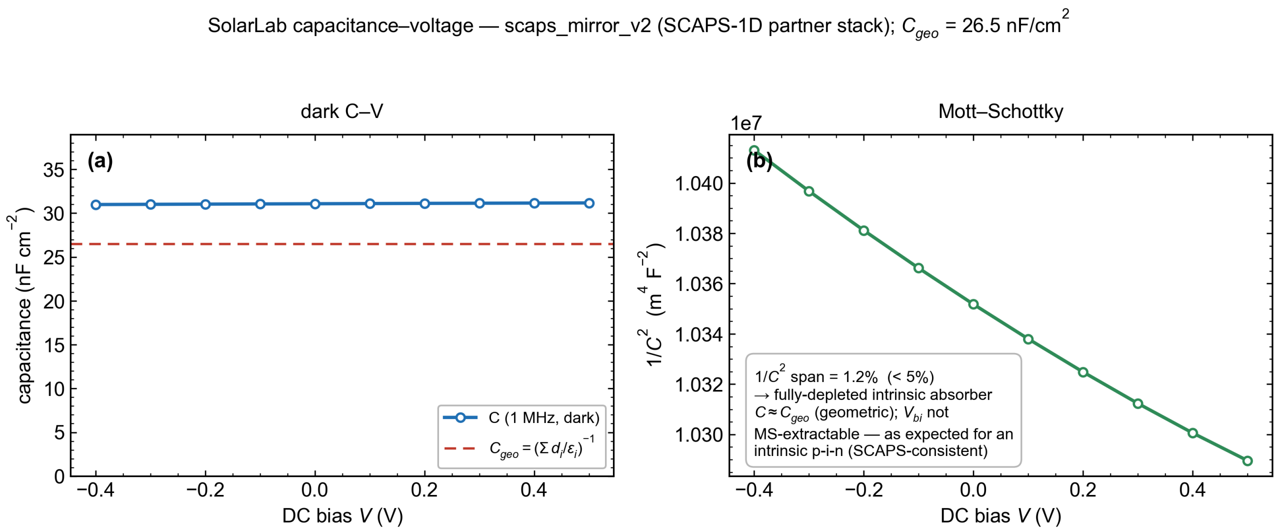

Figure 7: (a) Dark capacitance–voltage at 1 MHz with the geometric series capacitance C_{geo} for reference. (b) Mott–Schottky plot, $1/C^2$ versus bias.

The dark capacitance is essentially bias-independent at ≈ 31 nF cm^{-2}, close to the geometric series capacitance $C_{geo} = \left(\sum_i d_i/\varepsilon_i\right)^{-1} = 26.5$ nF cm^{-2}, and the Mott–Schottky ordinate $1/C^2$ varies by only 1.5% across the bias range. This is the diagnostic signature of a fully depleted intrinsic absorber: the depletion width does not vary with bias, the device behaves as a geometric parallel-plate capacitor, and a built-in voltage or doping density cannot be extracted from a Mott–Schottky fit. This is not a solver deficiency but the physically correct – and reference-consistent – result for an intrinsic p–i–n cell, and it is reported honestly here: the depletion-charge electrostatics are instead validated by the space-charge profile of Section 3.2 and by the agreement of the measured capacitance with C_{geo}.

4. Discussion

Taken together, the seven diagnostics verify the SolarLab physics across its four governing subsystems. The *electrostatics* are confirmed by the flat equilibrium Fermi level, the self-consistency of the band slope with the electric field, and the geometric capacitance. The *transport* is confirmed by the constant total current density (continuity to 10^{-3}) and the correct hand-off between electron and hole currents across the absorber. The *recombination* is resolved by mechanism and spatially localised to the hole-transport-layer interface, consistent across the recombination-profile and generation–recombination diagnostics. The *optics* are confirmed by the quantum-efficiency integral reproducing the short-circuit current to 2.6%. No internal inconsistency is observed at any operating point.

Two capabilities exercised by the reference SCAPS-family literature lie outside the present model and are stated explicitly to bound the verification. First, intra-band (thermionic-field-emission) tunnelling through a conduction-band spike is not modelled: the interface treatment is pure thermionic emission without a tunnelling transmission factor. Second, continuous bandgap grading is not yet supported, although the transport core already admits position-dependent band edges. Neither limitation affects the diagnostics reported here, which concern the standard planar device.

5. Conclusion

Seven independent depth-resolved and integral diagnostics, computed on the SCAPS-1D reference device and read directly from the solver's internal state, establish that the SolarLab drift–diffusion simulator is internally consistent and physically correct: it satisfies detailed balance at equilibrium, the qV quasi-Fermi splitting under bias, current continuity to a part in a thousand, optical–electrical current consistency to 2.6%, and the correct fully-depleted electrostatic signature of an intrinsic absorber. The verification is fully reproducible from the device configuration. On the basis of this evidence the solver's electrostatics, transport, recombination, and optics are sound on the reference device; the only excluded behaviours are two clearly delineated transport mechanisms not present in the current model.